

34870 Electroacoustics

Lab A — Analogy Circuits

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Abstract

All four systems of Lab A are modelled as lumped electrical networks and simulated with an AC sweep in KiCad 10 / ngspice 47. Mechanical parts use the *mobility* analogy (velocity $\leftrightarrow$ voltage, force $\leftrightarrow$ current), acoustic parts the *impedance* analogy (pressure $\leftrightarrow$ voltage, volume velocity $\leftrightarrow$ current). Couplings between domains are drawn with linear voltage-controlled sources (KiCad `GSOURCE/ESOURCE`). The physics is identical to the LTspice version the course uses; only the tool differs. Air: $\rho = 1.18 \text{ kg/m}^3$, $c = 344 \text{ m/s}$.

Conventions used throughout

	Mechanical (mobility)	Acoustical (impedance)	SPICE element
Across variable (node voltage)	velocity u [m/s]	pressure p [Pa]	V
Through variable (branch current)	force f [N]	volume velocity U [m ³ /s]	I
Mass / inertance	$M_M \rightarrow C = M_M$ (to ground)	$M_A \rightarrow L = M_A$	C / L
Compliance	$C_M \rightarrow L = C_M$	$C_A \rightarrow C = C_A$ (to ground)	L / C
Damping / loss	$R_M \rightarrow R = 1/R_M$	$R_A \rightarrow R = R_A$	R
Source	force $\rightarrow$ current source	$p \rightarrow$ V source, $U \rightarrow$ I source	I / V
Driving-point quantity	$Z_M = f/u = 1/V(u)$ for $I = 1 \text{ A}$	$Z_A = p/U$	

Table 1: Analogy conventions. In the mobility analogy a mechanical impedance appears as an *admittance* in the circuit, so $|Z_M|$ is read as $1/|V(u)|$ when the force source is 1 A.

Domain couplings (Lecture 4A): a vibrating area S gives $f = Sp$ and $U = Su$; a coil of length l in flux density B gives $f = Bli$ and $v_{\text{emf}} = Blu$. In the mobility analogy both are implemented with two voltage-controlled sources (one per direction), which is what LTspice's G/E and KiCad's `GSOURCE/ESOURCE` provide. All schematics, exported netlists and plotting scripts live in 5. Semester/Electroacoustics/Labs/Lab A/.

1 Part 1 — Mechanical system: dual-diaphragm loudspeaker

1.1 Model

The moving parts are two masses. M_{mvc} (coil, former and inner diaphragm) is driven by the force F and suspended to the frame by the spider $C_{msp} \parallel R_{msp}$; the outer diaphragm M_{md} is suspended by the surround $C_{msr} \parallel R_{msr}$ and connected to M_{mvc} through the joint $C_{md} \parallel R_{md}$. In the mobility analogy each mass is a node (its velocity), each mass a capacitor to ground, each spring-damper pair an inductor-resistor pair between the nodes they join.

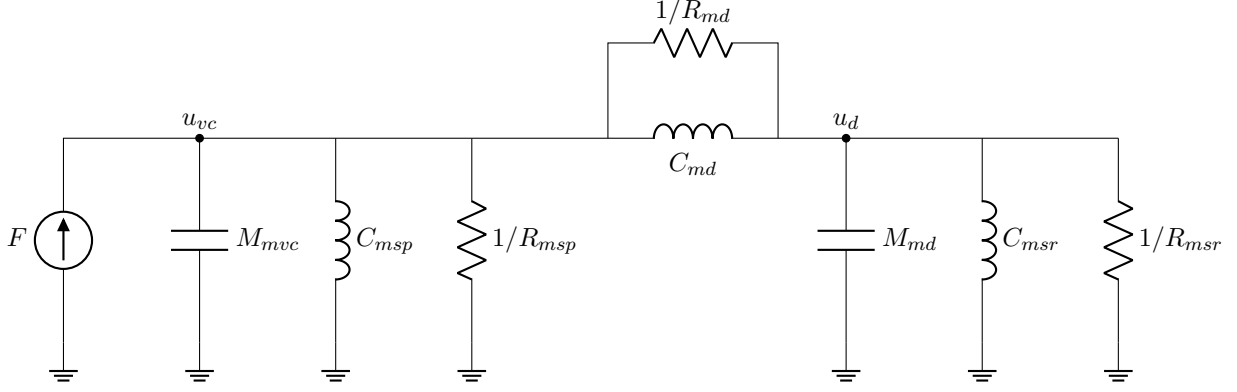


Figure 1: Part 1 mobility-analogy circuit. Node voltages are the velocities u_{vc} and u_d ; the current delivered by the source is the force F (1 N AC).

Physical	Value	SPICE element	Value
M_{md}	5 g	C (u_d -gnd)	5 mF
M_{mvc}	6 g	C (u_{vc} -gnd)	6 mF
C_{msp}	1.3 mm/N	L (u_{vc} -gnd)	1.3 mH
C_{msr}	2.7 mm/N	L (u_d -gnd)	2.7 mH
R_{msp}	0.5 N s/m	R (u_{vc} -gnd)	2
R_{msr}	0.22 N s/m	R (u_d -gnd)	4.545
C_{md} stiff / soft	10^{-10} / 3×10^{-6} m/N	L (u_{vc} - u_d)	100 pH / 3 μ H
R_{md} stiff / soft	5000 / 5 N s/m	R (u_{vc} - u_d)	0.2 m / 0.2

Table 2: Part 1 element values.

Hand estimate. With a stiff joint both masses move as one, $M_{tot} = 11$ g, on the two suspensions in parallel (stiffnesses add): $1/C_{tot} = 1/C_{msp} + 1/C_{msr}$, $C_{tot} = 0.878$ mm/N, so $f_0 = 1/(2\pi\sqrt{M_{tot}C_{tot}}) \approx 51.2$ Hz.

1.2 Results and discussion

Case	Feature	f	Value
Stiff	single resonance, $u_{vc} = u_d$	51.3 Hz	$1.389 \text{ m/(s N)} = 1/R_{tot}$
Stiff	u_d/u_{vc} over the whole band	—	1.000 (0.998 at 10 kHz)
Soft	first resonance (masses in phase)	51.3 Hz	1.388 (u_{vc}), 1.389 (u_d)
Soft	u_{vc} dip (absorber anti-resonance)	1288 Hz	0.0030 m/(s N)
Soft	u_d maximum near the link resonance	1738 Hz	0.0493
Soft	second resonance (masses in anti-phase)	1799 Hz	u_{vc} 0.0415
Soft	u_{vc}/u_d at 10 kHz	—	42 (outer diaphragm 32 dB below the coil)

Table 3: Part 1a extracted numbers.

a) Velocity response (Fig. 2, ratio in Fig. 3). *Stiff link.* $C_{md} = 10^{-10}$ m/N is a rigid rod at every frequency of interest (its own resonance with the 5 g mass would sit at 225 kHz), so both masses move as one 11 g body on the two suspensions in parallel. There is one resonance at 51.3 Hz (hand estimate 51.2 Hz), whose peak height $1/R_{tot} = 1/(0.5 + 0.22) = 1.389 \text{ m/(s N)}$ is set purely by the two dampers. Below it the suspension compliance controls the motion ($u \propto f$, $+90^\circ$), above it the mass controls ($u \propto 1/f$, -90° , -6 dB per octave). The two curves are identical.

Soft link. Below about 300 Hz the link is still stiff compared with the inertia of the masses, so the low resonance is unchanged (51.3 Hz, same peak). Around 1.3 kHz the outer diaphragm

on its soft spring becomes a *tuned absorber*: $f = 1/(2\pi\sqrt{M_{md}C_{md}}) = 1300$ Hz by hand, 1288 Hz simulated. It resonates against the coil, the reaction force through C_{md} cancels the drive and u_{vc} dips by about 30 dB while u_d is still large. Just above, at 1.7 kHz to 1.8 kHz, the two masses swing in anti-phase (second resonance; the undamped two-mass estimate with the reduced mass $M_{md}M_{mvc}/(M_{md} + M_{mvc})$ gives 1760 Hz, simulated 1799 Hz) and the phase of u_d flips through -180° . Above that the soft link can no longer transmit force ($u_d/u_{vc} \propto 1/\omega^2$ through a compliance), so u_d rolls off 12 dB per octave faster than u_{vc} and the coil ends up driving only its own 6 g. This is the Lowther dual-diaphragm trick: the large cone works up to roughly 1.5 kHz, then decouples, and the light inner cone alone carries the top octaves without the big cone's mass and break-up.

Part 1a - velocity response to $F = 1$ N

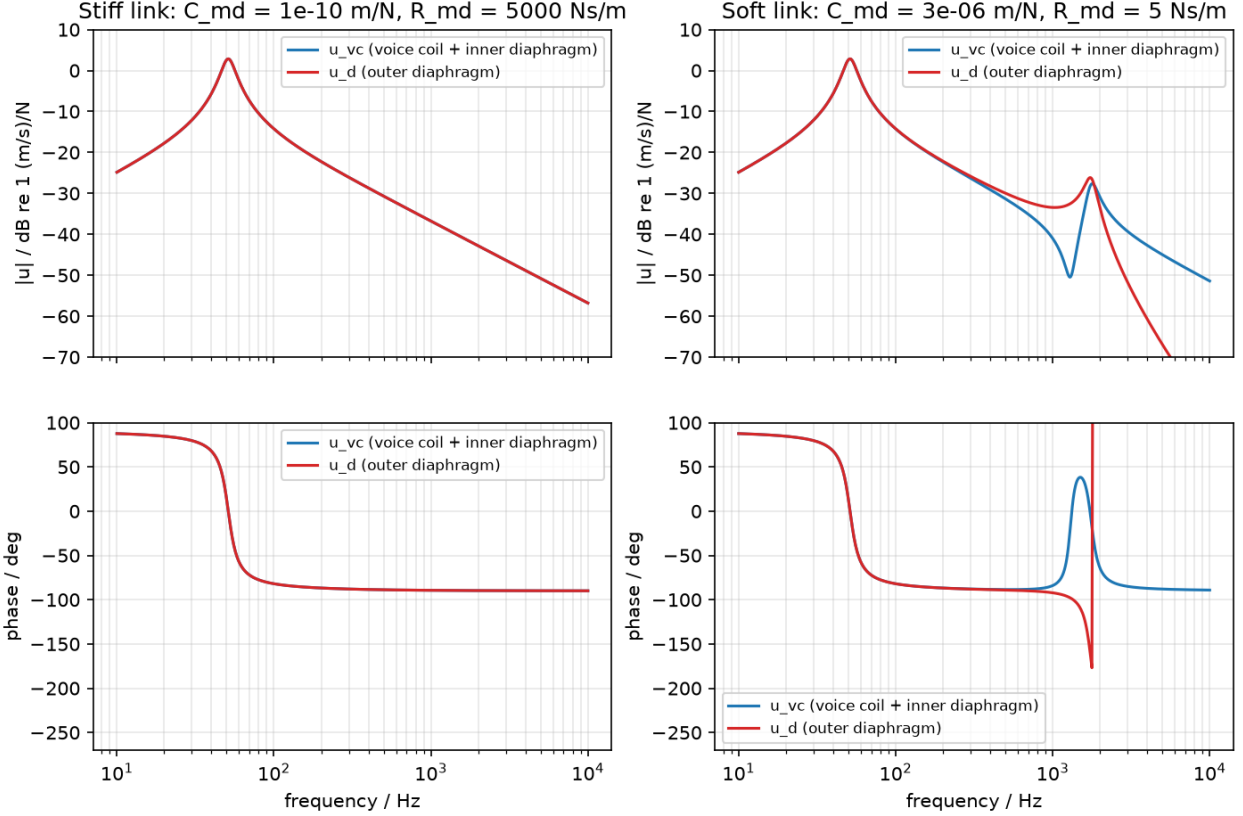


Figure 2: Part 1a: velocity of the coil/inner diaphragm u_{vc} and of the outer diaphragm u_d per newton of force ($F = 1$ N AC), magnitude and phase, stiff link (left) and soft link (right), 10 Hz to 10 000 Hz.

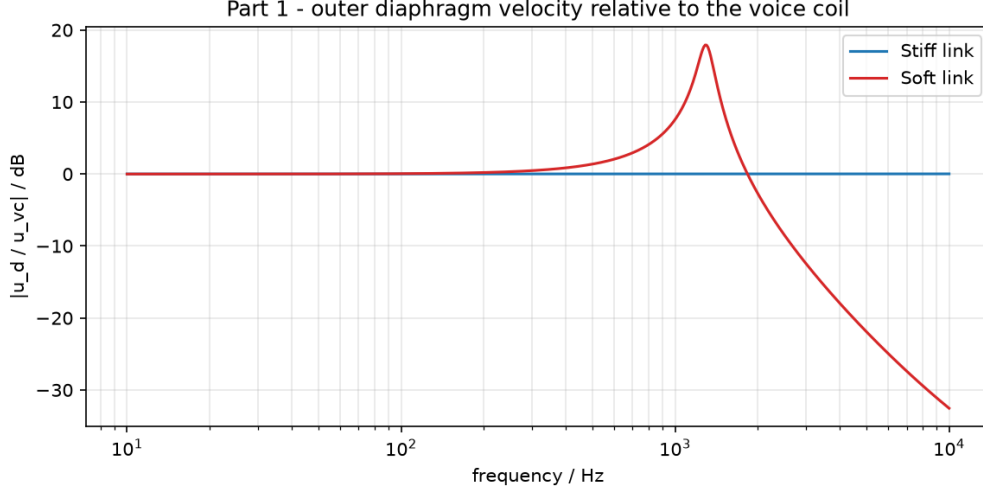


Figure 3: Part 1a: ratio u_d/u_{vc} for the two links — unity for the stiff link, the mechanical crossover for the soft link.

Case	Feature	f	Z_M [N s/m]
both	compliance asymptote	10 Hz	$0.72 - j17.4$ ($\approx 1/(\omega C_{tot}) = 18.1$)
both	resonance (minimum)	51.3 Hz	$0.720 = R_{tot}$, phase 0°
Stiff	mass asymptote	10 kHz	$0.74 + j692$ ($= \omega M_{tot}$, 11 g)
Soft	anti-resonance (maximum)	1288 Hz	334
Soft	second resonance (minimum)	1799 Hz	24.1
Soft	mass asymptote	10 kHz	$5.7 + j372$ ($= \omega M_{mvc}$, only the 6 g coil)

Table 4: Part 1b extracted numbers.

b) Mechanical input impedance $Z_M = F/u_{vc}$ (Fig. 4). The impedance is the mirror image of u_{vc} (with a 1 A source it is literally $1/V(u_{vc})$). Below resonance the suspension dominates: $|Z| \propto 1/f$ with phase -90° , the *compliance line*. At resonance the reactances cancel and the floor is the pure damping $R_{tot} = 0.72$ N s/m at phase 0° . Above resonance the mass dominates: $|Z| \propto f$ with phase $+90^\circ$, the *mass line*. For the stiff link the mass line follows 11 g all the way to 10 kHz. For the soft link it follows 11 g up to about 1 kHz, then the absorber produces a *maximum* (anti-resonance, 334 N s/m: the outer diaphragm pumps force back into the coil), then a *minimum* at 1.8 kHz (anti-phase resonance), and finally the line continues at only 6 g because the outer diaphragm has fallen off. The residual real part of 5.7 N s/m at 10 kHz is the power still dissipated in R_{md} by the relative motion of the two masses.

Part 1b - mechanical input impedance $Z_M = F / u_{vc}$

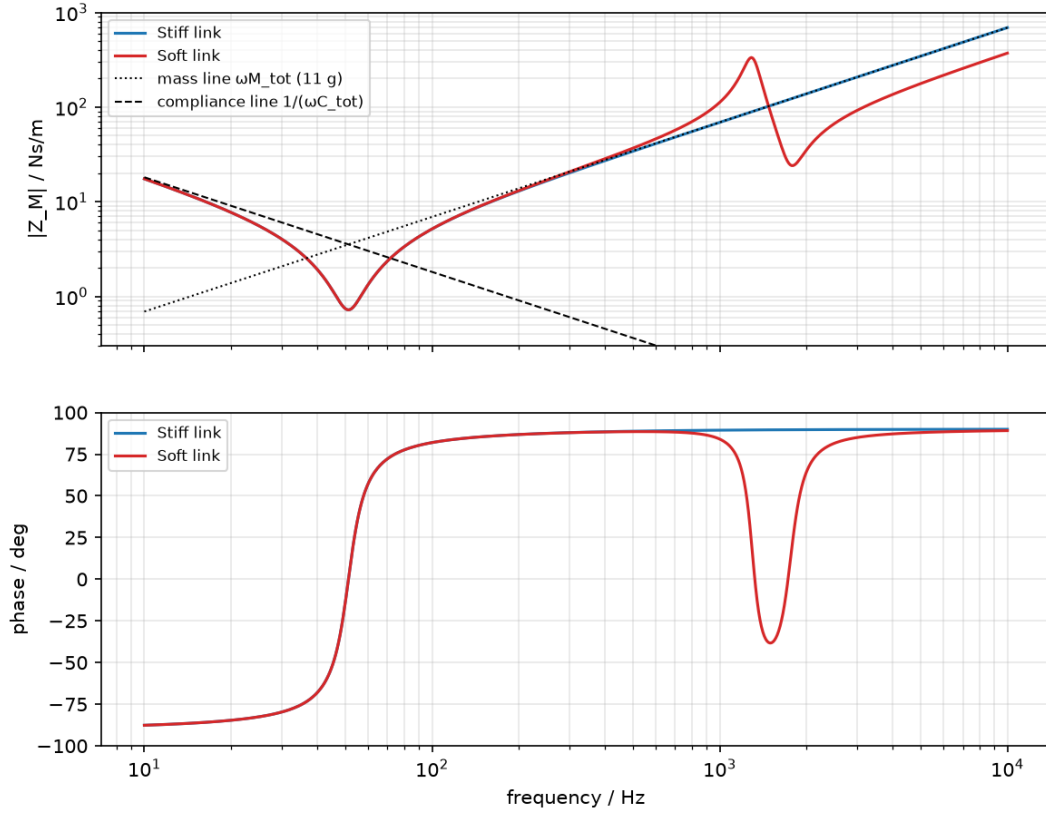


Figure 4: Part 1b: mechanical input impedance seen by the force, magnitude and phase, stiff and soft link.

Modelling choices and validity. No acoustic load (as instructed), so all damping is the three mechanical dampers and the response is that of an ideal lumped two-mass system; a real cone would add break-up modes above a few kilohertz. The frequencies are read from a 200-points-per-decade sweep, accurate to about 0.5%.

2 Part 2 — Acoustic system: car silencer

2.1 Model

Elements 1, 3, 5, 7 are narrow tubes (radius 2 mm); each is an acoustic mass $M_A = \rho l / S$ in series with the given loss $R_A = 25 \times 10^3 \text{ Pa s / m}^3$. Elements 2, 4, 6 are expansion chambers; each is a compliance $C_A = V / (\rho c^2)$ from its node to the reference pressure (ground). The result is a ladder network (a low-pass filter), driven either by a pressure source (voltage) or a volume-velocity source (current).

Element	r [mm]	l [mm]	S or V	M_A [kg/m ⁴]	C_A [m ⁵ /N]	R_A
1 tube	2	25	$1.257 \times 10^{-5} \text{ m}^2$	2348	—	25×10^3
2 chamber	10	50	15.7 cm^3	—	1.125×10^{-10}	—
3 tube	2	80	$1.257 \times 10^{-5} \text{ m}^2$	7512	—	25×10^3
4 chamber	20	80	100.5 cm^3	—	7.200×10^{-10}	—
5 tube	2	100	$1.257 \times 10^{-5} \text{ m}^2$	9390	—	25×10^3
6 chamber	15	60	42.4 cm^3	—	3.037×10^{-10}	—
7 tube	2	40	$1.257 \times 10^{-5} \text{ m}^2$	3756	—	25×10^3

Table 5: Silencer element values (effective lengths as given in the lab).

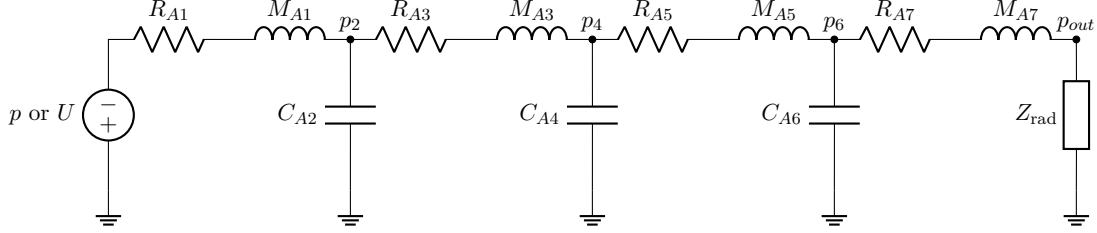


Figure 5: Silencer ladder. For a)–c) the outlet is an ideal open end ($Z_{\text{rad}} = 0$, $p_{\text{out}} = 0$); for d)–e) the tube-end radiation network is inserted.

Radiation impedance at the outlet (d, e). For an unflanged tube end of radius a (piston in a long tube, Lecture 3 §8b) the fitted network is $Z_{\text{rad}} = j\omega M_{A1} \parallel [R_{A2} + (R_{A1} \parallel C_{A1})]$ with $M_{A1} = 0.6133\rho/(\pi a)$, $C_{A1} = 0.55\pi^2 a^3/(\rho c^2)$, $R_{A1} = 0.5045\rho c/(\pi a^2)$, $R_{A2} = \rho c/(\pi a^2)$. With $a = 2 \text{ mm}$: $M_{A1} = 115.2 \text{ kg/m}^4$, $C_{A1} = 3.11 \times 10^{-13} \text{ m}^5/\text{N}$, $R_{A1} = 1.63 \times 10^7$, $R_{A2} = 3.23 \times 10^7 \text{ Pa s/m}^3$. Because the given pipe lengths are already *effective* lengths, the end correction $0.6133a = 1.23 \text{ mm}$ is subtracted from pipe 7 when the radiation mass is present, so it is not counted twice. Over the whole band $ka \leq 0.037$, so the radiation load is essentially the mass M_{A1} and the near-field pressure at the opening is $p_{\text{out}} \approx j\omega M_{A1} U_{\text{out}}$.

2.2 Results and discussion

Hand estimates of the natural frequencies from single mass–compliance pairs are only a rough guide, since the ladder is a three-degree-of-freedom system whose modes mix neighbouring elements. The most useful estimates take a chamber with the two tubes on either side of it *in parallel*: C_{A2} with $M_{A1} \parallel M_{A3}$ gives 355 Hz, C_{A4} with $M_{A3} \parallel M_{A5}$ 92 Hz, C_{A6} with $M_{A5} \parallel M_{A7}$ 176 Hz; the first and last coincide with simulated peaks. All four masses in series ($23\,006 \text{ kg/m}^4$) against all three compliances in parallel ($1.136 \times 10^{-9} \text{ m}^5/\text{N}$) give 31 Hz.

Feature	f	$ U_{\text{out}}/p_{\text{in}} $ [dB re $1 \text{ m}^3/(\text{s Pa})$]
10 Hz	—	−123.0
dip	42.2 Hz	−132.0
peak	77.1 Hz	−98.9
dip	142.9 Hz	−143.9
peak	179.9 Hz	−112.3
dip	309.7 Hz	−170.3
peak	356.5 Hz	−141.6
1 kHz	—	−253.9

Table 6: Part 2a: output volume velocity of pipe 7 for a 1 Pa pressure source at the inlet.

a) Pressure source (Fig. 6a). The output is $U_{out} = (p_{in}/Z_{in}) \cdot (U_{out}/U_{in})$, so with a pressure source the low-frequency level is set by the *input impedance of the whole ladder* (Fig. 7b). Below about 1 Hz that would be the four series losses $\sum R_A = 10^5 \text{ Pa s/m}^3$ (−100 dB), but already at 10 Hz the series air masses dominate: $\omega \sum M_A = 2\pi \cdot 10 \cdot 23006 = 1.45 \times 10^6 \gg 10^5$, hence $|Z_{in}| \approx 1.5 \times 10^6$ and −123 dB. The three peaks (77.1, 179.9, 356.5 Hz) are the *minima* of Z_{in} , i.e. the series resonances: with the inlet held at a fixed pressure the natural frequencies are those of the ladder with a short-circuited ($p = 0$) inlet. The dips are where U_{out} passes through zero between two poles, close to the Z_{in} maxima (47, 176, 191 Hz) and to the M_{A1} – C_{A2} pair (310 Hz). Above the last resonance the response falls at about 18 dB per octave per chamber section (a ladder of three L–C low-pass sections), from −142 dB at 357 Hz to −254 dB at 1 kHz. The losses R_A only round off the peaks: at 100 Hz $\omega M_{A3} = 4.7 \times 10^6 \gg 2.5 \times 10^4$, so the Q-factors are in the hundreds and the peaks are very sharp — sharp enough that the sweep needs about 2000 points per decade to resolve their height (200 points per decade reads the 77 Hz peak 2 dB and the 357 Hz peak 8 dB too low).

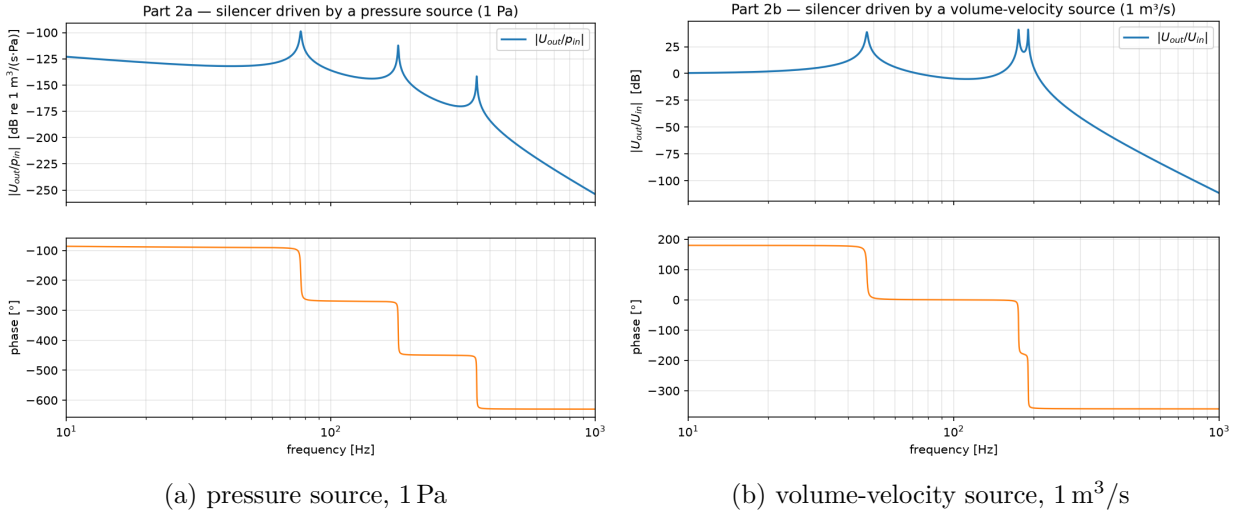


Figure 6: Part 2a/b: volume velocity in the exit pipe, $|I(V_{s7})|$, magnitude and phase, 10 Hz to 1000 Hz.

Feature	f	$ U_{out}/U_{in} $ [dB]
10 Hz	—	+0.45 ($\rightarrow 0$ as $f \rightarrow 0$)
peak	47.1 Hz	+38.6
dip	112.1 Hz	−5.2
peak	175.6 Hz	+40.9
dip	183.7 Hz	+20.1
peak	190.8 Hz	+41.1
1 kHz	—	−111.4

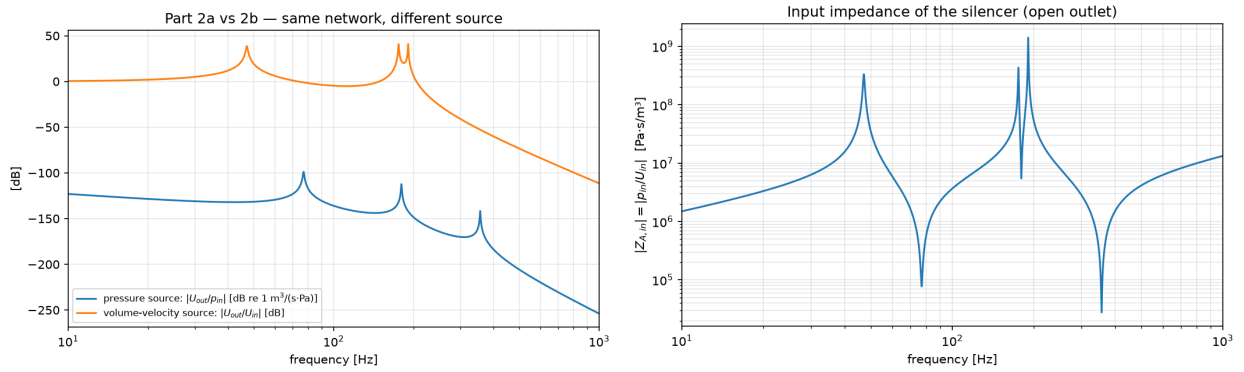
Table 7: Part 2b: transfer U_{out}/U_{in} for the volume-velocity source.

b) Volume-velocity source (Fig. 6b) and the difference to a). Transmission loss $TL = -20 \log |U_{out}/U_{in}|$: −1.9 dB at 20 Hz, −19.3 dB at 50 Hz, +4.7 at 100, −0.5 at 150, −6.1 at 200, +40.9 at 300, +73.4 at 500, +92.2 at 700 and +111.4 dB at 1 kHz.

With an ideal volume-velocity source the flow is forced into the inlet whatever pressure it takes. At low frequency the chambers cannot store flow ($1/(\omega C_A)$ is huge), so everything that goes in comes out: 0 dB, independent of the series masses and losses. The transfer U_{out}/U_{in} is a property of the network alone; the source only decides which natural frequencies are excited. A volume-velocity source is an *open circuit* at the inlet, so the peaks sit at the open-inlet natural frequencies, the

maxima of Z_{in} (47.1, 175.6 and 190.8 Hz) — exactly where the pressure-source response has its dips — while the pressure-source peaks (Z_{in} minima) do not appear at all (overlay in Fig. 7a). At those frequencies the ladder resonates and the silencer *amplifies* the flow by about 40 dB, limited only by the pipe losses. A real engine exhaust is somewhere between the two ideal sources (finite source impedance), which is why a silencer design has to know its source impedance. Above 200 Hz the chambers shunt the flow and the mass of the following pipe blocks it; the roll-off is the same 18 dB per octave per section, reaching 111 dB of transmission loss at 1 kHz. The second and third open-inlet modes are close neighbours (176 and 191 Hz) separated by a +20 dB saddle.

A modelling trap worth recording. The ideal current source must not be loaded. At these anti-resonances $|Z_{in}|$ reaches 2.9×10^8 Pa s/m³ at 47 Hz and 1.5×10^9 at 191 Hz, so a 100 M “DC-leak” resistor across the source — a common SPICE habit — takes most of the source flow exactly at the peaks: it lowers them to +25.8 and +26.6 dB and merges the upper two into one hump. The pipe inductors already provide the DC path, so no leak is needed (the sheets use 1 T); LTspice with an unloaded source gives the same three peaks.



(a) U_{out} for both sources, normalised

(b) input impedance $Z_{in} = p_{in}/U_{in}$ of the ladder

Figure 7: Part 2: why the two sources excite different resonances — the pressure source peaks at the minima of Z_{in} , the volume-velocity source at its maxima.

Pipe	Level at 1 kHz	Peaks / dips
1	0 dB everywhere	(it <i>is</i> the source current)
3	−30.2 dB	+21.0 dB at 47 Hz, +34.6 at 176 Hz, +45.6 at 191 Hz; dip −22 dB at 51 Hz
5	−78.6 dB	+37.7 at 47 Hz, +32.7 at 176 Hz, +37.2 at 191 Hz; dip −43 dB at 149 Hz
7	−111.4 dB	+38.6 at 47 Hz, +40.9 at 176 Hz, +41.1 at 191 Hz

Table 8: Part 2c: volume velocity in pipes 1, 3, 5, 7 relative to U_{in} .

c) Volume velocity in each pipe, volume-velocity source (Fig. 8). $I(V_{s1}) = U_{in}$ by definition, a flat 0 dB line. Each expansion chamber acts as an acoustic low-pass: chamber 2 shunts part of U_1 so $U_3 < U_1$, chamber 4 takes another bite, chamber 6 the last. At 1 kHz the cumulative attenuation is 30, 79 and 112 dB after one, two and three chambers, i.e. each section contributes 30–40 dB there. The dips in U_3 (51 Hz) and U_5 (149 Hz) are anti-resonances where the downstream part of the ladder presents a very high impedance, so almost no flow enters that pipe and the chamber before it takes it all; pipe 7 has no dip because nothing follows it. The resonance peaks appear in every pipe but grow towards the outlet: energy stored in the resonating chambers is released into the pipe after them.

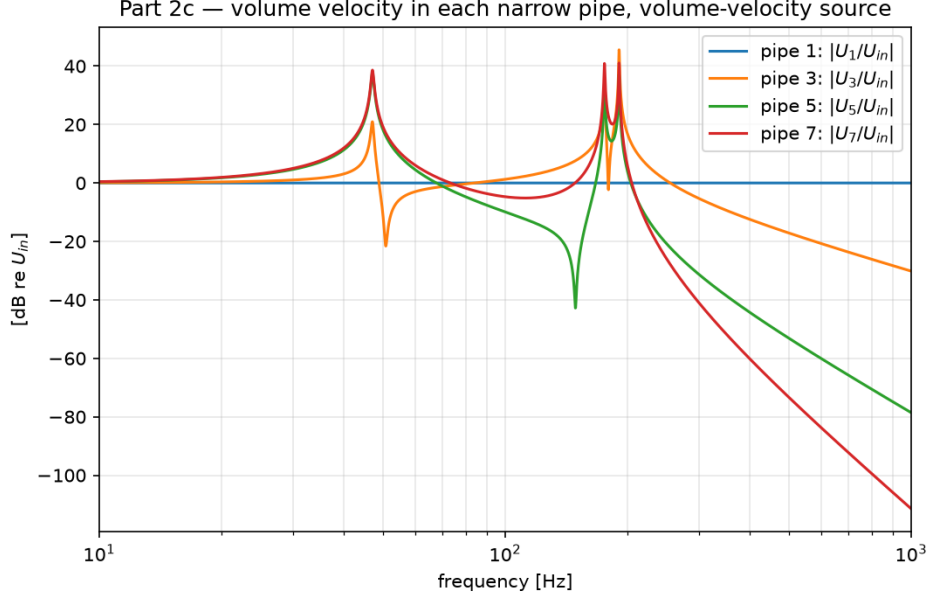


Figure 8: Part 2c: $|I(V_{s1})|$, $|I(V_{s3})|$, $|I(V_{s5})|$, $|I(V_{s7})|$ in dB re $U_{in} = 1 \text{ m}^3/\text{s}$.

d) Near-field pressure at the opening (Fig. 9a). With the radiation network in place, $p_{open} = V(p_{out})$: 77.6 dB re 1 Pa at 10 Hz (7.6 kPa per m^3/s), 92.5 dB at 100 Hz, 5.7 dB at 1 kHz. The radiation impedance of the 2 mm tube end is essentially a *pure mass* over the whole band: $ka = 0.037$ at 1 kHz, $|Z_{rad}| = 7.24 \times 10^5 \text{ Pa s/m}^3$ with a real part of only 1.08×10^4 (1.5 %); the resistive $R_{A2} = 3.2 \times 10^7$ is only reached near $ka = 1$ at 27 kHz. Consequently $p_{open} \approx j\omega M_{A1} U_{out}$: the dashed curve $\omega M_{A1} |U_{out}|$ lies on top of the simulated pressure, and the pressure curve is simply the U_{out} curve of b) tilted by +6 dB per octave, with the same peaks at 47, 176 and 191 Hz. Inserting the radiation load changes U_{out} by less than 0.01 dB: $|Z_{rad}| \leq 7 \times 10^5$ against a pipe-7 impedance ωM_{A7} of 2.3×10^5 at 10 Hz and 2.3×10^7 at 1 kHz. The silencer does not “see” the outside air; the ideal open end used in a)–c) is an excellent approximation.

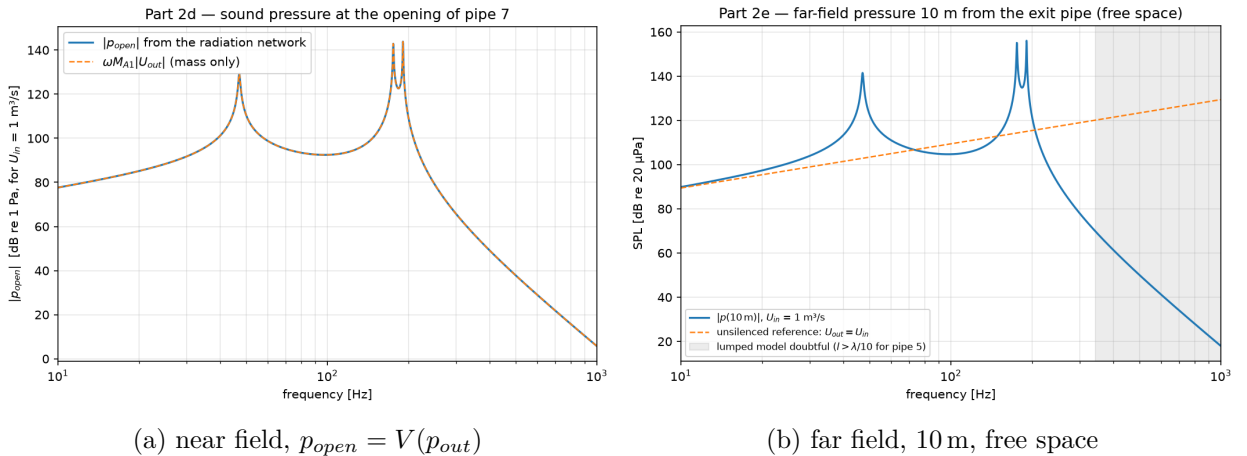


Figure 9: Part 2d/e: sound pressure at the exit and at 10 m for $U_{in} = 1 \text{ m}^3/\text{s}$ (a realistic $10^{-3} \text{ m}^3/\text{s}$ is 60 dB lower). The grey band marks where the lumped model is no longer valid.

e) Pressure at 10 m (Fig. 9b) and validity. The pipe end radiates into free space (4π) as a point source: $p(r) = j\omega\rho U_{out} e^{-jkr}/(4\pi r)$, so $|p| = \rho f |U_{out}|/(2r)$, computed from $I(V_{s7})$ of the radiation sheet.

f	SPL with silencer [dB re 20 μ Pa]	unsilenced ($U_{out} = U_{in}$)
20 Hz	97.4	95.4
50 Hz	122.6	103.4
100 Hz	104.7	109.4
200 Hz	121.5	115.4
500 Hz	50.0	123.4
1 kHz	17.9	129.4

Table 9: Part 2e: sound pressure level at 10 m for $U_{in} = 1 \text{ m}^3/\text{s}$.

The far-field pressure of a monopole rises 6 dB per octave for constant U (the dashed reference), so the radiated SPL is the U_{out} transfer of b) plus that tilt. The silencer helps only above about 205 Hz (roughly 73 dB at 500 Hz, 111 dB at 1 kHz) and *hurts* by up to about 41 dB at its three resonances when driven by an ideal volume-velocity source; the finite source impedance of a real engine is what tames those peaks.

Validity at higher frequencies. Lumped elements require $l < \lambda/10$. Pipe 5 (100 mm) meets this only below 344 Hz, pipe 3 and chamber 4 (80 mm) below 430 Hz, chamber 6 below 573 Hz, chamber 2 below 688 Hz, pipe 7 below 860 Hz. Above roughly 350–450 Hz the pipes and chambers must be treated as transmission lines (ABCD two-ports, Lecture 3 §4); the smooth lumped roll-off then becomes a sequence of pass and stop bands (the first half-wave resonance of the 100 mm pipe is $c/2l = 1.7 \text{ kHz}$, of the 80 mm chamber 2.15 kHz). The 111 dB of transmission loss at 1 kHz is therefore not to be trusted. The plane-wave assumption across the chamber radius holds ($ka = 1$ at 2.7 kHz for the 20 mm chamber); the point-source assumption for the 2 mm exit is excellent ($ka \ll 1$ to tens of kilohertz) and $r = 10 \text{ m} \gg \lambda$ except at the lowest frequencies. Free-space radiation ignores the car body and the road (a tailpipe near the ground sees a half-space, +6 dB, plus reflections); the constant loss R_A is crude, since viscous losses in a 2 mm pipe grow with $\sqrt{f}$; the walls are assumed rigid, and hot exhaust gas has a different ρ and c than the 20 °C air assumed here.

3 Part 3 — Coupled mechanical–acoustic system

3.1 Model

Each diaphragm now radiates from both sides into an infinite baffle. A piston of area S sees, per side, the baffled-piston radiation impedance (Lecture 3 §8b)

$$Z_{\text{rad}}(a) = j\omega M_{A1} \parallel [R_{A2} + (R_{A1} \parallel C_{A1})],$$

$$M_{A1} = \frac{8\rho}{3\pi^2 a}, \quad R_{A1} = 0.441 \frac{\rho c}{\pi a^2}, \quad R_{A2} = \frac{\rho c}{\pi a^2}, \quad C_{A1} = 5.94 \frac{a^3}{\rho c^2},$$

and both sides together give $2Z_{\text{rad}}$. The coupling to the mobility circuit uses $U = Su$ and $f = Sp$: a VCCS with gain S injects the volume velocity into the acoustic sub-network, and a second VCCS with gain S draws the reaction force Sp from the velocity node. Seen from the mechanical side this is the load $Z_{M,A} = S^2 \cdot 2Z_{\text{rad}}$ (an admittance $S^2 \cdot 2Z_{\text{rad}}$ in the mobility circuit).

Diaphragm	S [cm ²]	a [mm]	M_{A1} [kg/m ⁴]	R_{A1} [Pa s/m ³]	R_{A2}	C_{A1} [m ⁵ /N]	$2S^2 M_{A1}$ [g]	$2S^2 R_{A2}$ [N s/m]
inner	60	43.7	7.295	29835	67653	3.55×10^{-9}	0.525	4.87
outer	210	81.8	3.900	8524	19330	2.33×10^{-8}	3.44	17.05

Table 10: Radiation-impedance constants per side, and the resulting added mass (low ka) and added resistance (high ka) for both sides. $ka = 1$ at 1253 Hz (inner) and 670 Hz (outer).

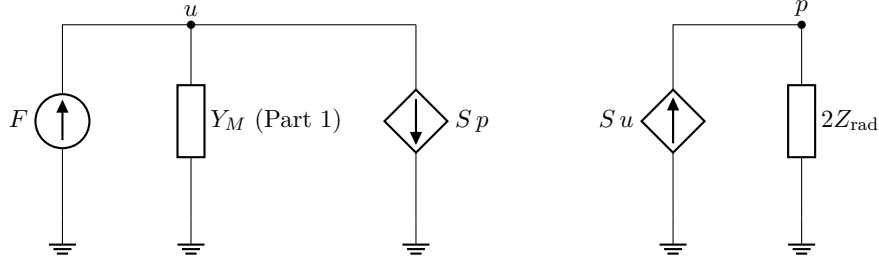


Figure 10: Coupling block used for each diaphragm in Part 3 (mobility on the left, acoustic impedance analogy on the right). The left controlled source senses p and draws the reaction force, the right one senses u and injects the volume velocity. The pressure on *one* side of the diaphragm is $p/2$.

3.2 Results and discussion

Sign check of the coupling. With the air load the resonance moves *down* (51.3 to 43.7 Hz; hand: $51.2\sqrt{11/(11 + 0.525 + 3.44)} = 43.9$ Hz), the peak velocity *drops* (1.389 to 1.314 m/(s N)) and the real part of Z_M at 10 kHz rises from 0.74 to 22.8 N s/m ($= 0.72 + 4.87 + 17.05 = 22.6$). A wrong polarity of either controlled source would have raised the peak or made the circuit unstable, so the coupling is a passive load as it must be.

Case	Feature	Part 1 (no air)	Part 3 (with air)
Stiff	resonance	51.3 Hz, 1.389	43.7 Hz, 1.314
Stiff	$ Z_M $ minimum	0.720	0.761
Stiff	Z_M at 10 kHz	$0.74 + j692$	$22.8 + j693$
Soft	first resonance	51.3 Hz, 1.388	43.7 Hz, 1.314
Soft	u_{vc} dip / $ Z_M $ maximum	1288 Hz, 334	1023 Hz, 129
Soft	second resonance (u_{vc} peak, $ Z_M $ min)	1799 Hz, 0.0415, 24.1	1718 Hz, 0.0264, 37.9
Soft	u_d maximum	1738 Hz, 0.0493	1549 Hz, 0.0257
Soft	Z_M at 10 kHz	$5.7 + j372$	$10.6 + j372$
Soft	u_{vc}/u_d at 10 kHz	42	43

Table 11: Part 3a: Part 1 versus Part 3. Velocities in m/(s N), impedances in N s/m.

a) Velocities and impedance with and without air (Figs. 11, 12). *Low frequency, $ka \ll 1$.* The radiation impedance is almost purely mass-like ($j\omega M_{A1}$ dominates the parallel network): the air adds 0.5 g on the inner and 3.4 g on the outer diaphragm to the 11 g, so the resonance drops to 43.7 Hz for both links. The radiation resistance there is tiny ($\propto (ka)^2$), so the peak height is nearly unchanged; the drop from 1.389 to 1.314 is mostly the Q change from the extra mass at fixed R .

Around $ka \approx 1$ (670 Hz outer, 1253 Hz inner) the network turns resistive. For the soft link the absorber frequency of the outer diaphragm drops from 1288 to 1023 Hz (its mass grew from 5 to 8.4 g: $1/(2\pi\sqrt{8.44\text{ g} \cdot 3\mu\text{m/N}}) = 1000$ Hz) and, more importantly, its 17 N s/m of radiation resistance damps the absorber: the u_{vc} dip is 8 dB shallower and the anti-resonance in Z_M falls from 334 to 129 N s/m. The second resonance is likewise lower in frequency and about 4 dB lower in amplitude.

High frequency, $ka \gg 1$. The load is a pure resistance $\rho c S^2$ per side. Stiff link: 21.9 N s/m in series with the 11 g mass line, invisible in $|Z_M|$ (692 N s/m reactive) but it is the entire radiated power. Soft link: only the inner diaphragm is still moving, so only its 4.87 N s/m shows up ($10.6 = 5.7 + 4.9$).

Effect of the radiation impedance in one sentence: a frequency-dependent load that is extra mass at low frequency (lowers every resonance) and extra damping at high frequency (flattens the absorber anti-resonance and sets the radiated power); on the outer diaphragm it is about seven times larger than on the inner one.

Part 3a - velocities with and without the radiation load ($F = 1$ N)

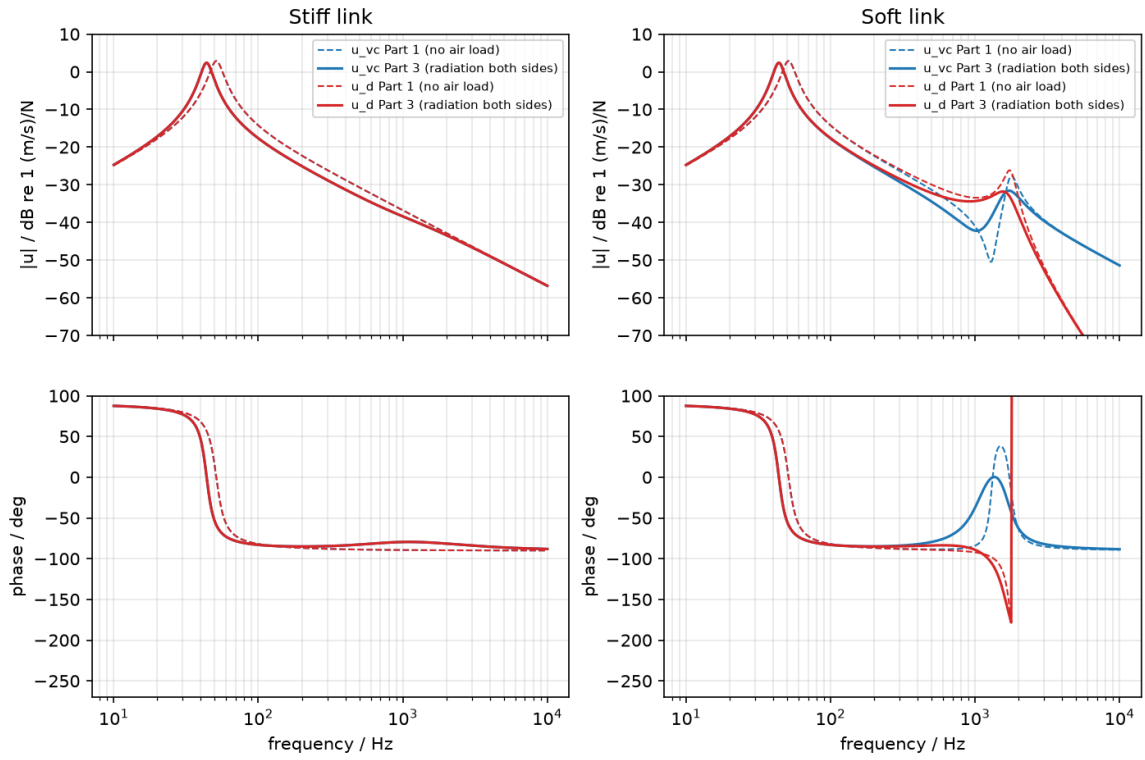


Figure 11: Part 3a: u_{vc} and u_d per newton with (solid) and without (dashed, Part 1) radiation loading, stiff and soft link.

Part 3a - mechanical input impedance $Z_M = F / u_{vc}$, with and without air load

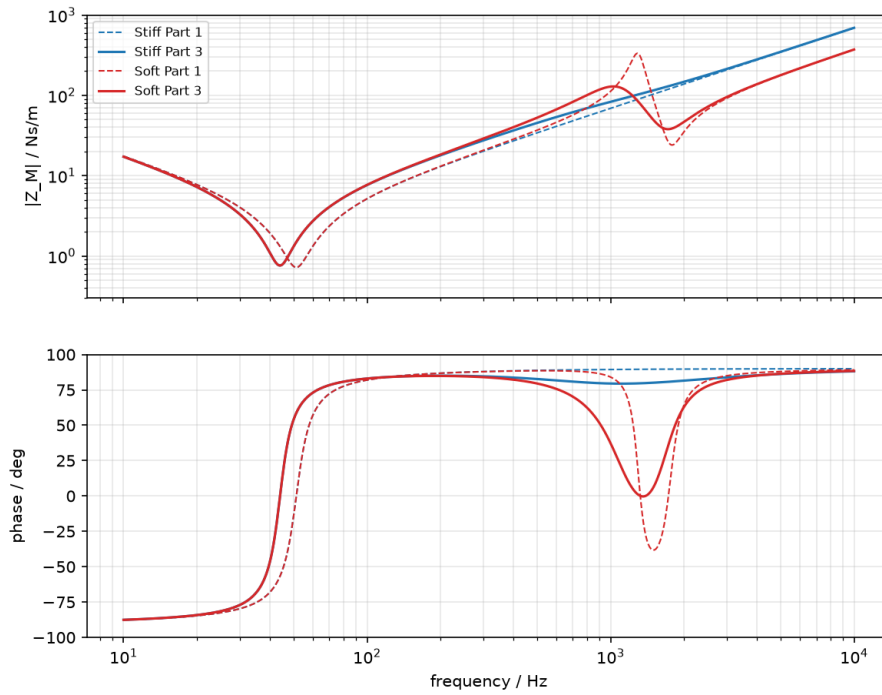


Figure 12: Part 3a: mechanical input impedance with (solid) and without (dashed) radiation loading.

b) Sound pressure (Fig. 13). The pressure in front of one face is $V(p_x)/2$, since the p node carries the sum of both faces. The far-field pressure on axis at 1 m in the baffle half-space is $p = j\omega\rho(S_i u_{vc} + S_o u_d)/(2\pi \cdot 1 \text{ m})$, computed from the node voltages.

Case	Quantity	Value [Pa/N]
both	p_{front} inner / outer at the 44 Hz resonance	15.9 / 29.8
Stiff	p_{front} plateau 100 Hz to 1000 Hz, inner / outer	$\approx 3 / 5.6$ (10 / 15 dB)
Stiff	p_{front} at 10 kHz, both	0.59
Soft	p_{front} inner: 2nd peak / 10 kHz	10.0 at 1738 Hz / 1.10
Soft	p_{front} outer: 2nd peak / 10 kHz	10.9 at 1567 Hz / 0.03
both	far field peak at 44 Hz	1.85
Stiff	far field 100 Hz to 10 000 Hz	flat 0.35–0.46
Soft	far field	0.35 plateau, +8 dB hump at 1.6 kHz (0.87), –18 dB dip near 3 kHz (0.12), 0.18 at 10 kHz

Table 12: Part 3b extracted numbers.

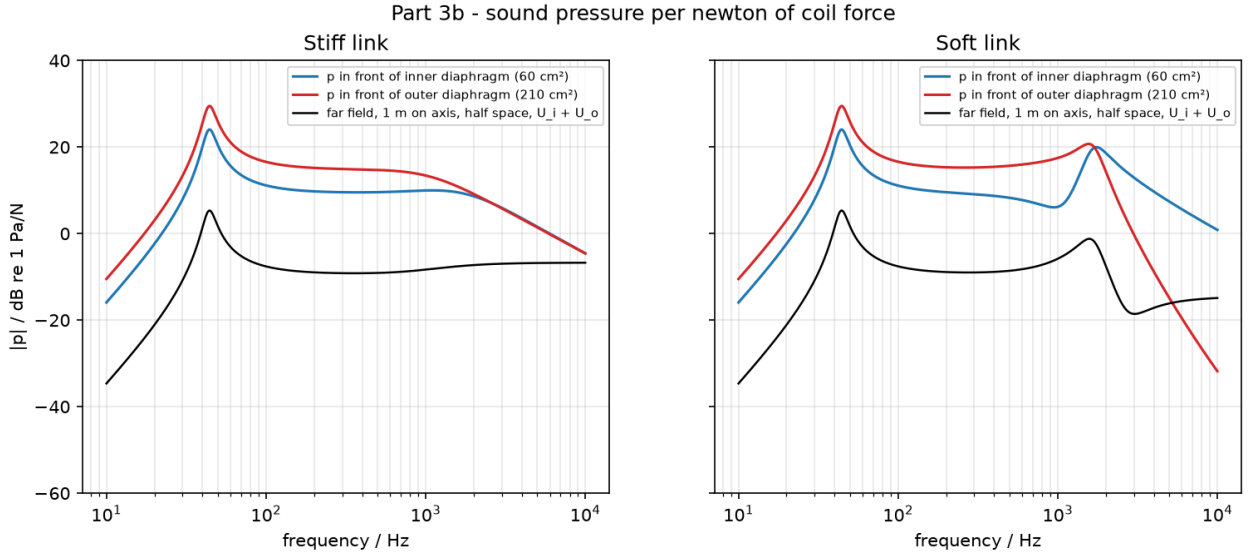


Figure 13: Part 3b: pressure in front of each diaphragm and on-axis far-field pressure at 1 m (half space, $U_i + U_o$), per newton of coil force.

The near-field pressure per face is $Z_{rad}(f) S u$. Below $ka = 1$ the load is mass-like, so $p_{front} \approx \omega M_{A1} S u$, which with $u \propto 1/f$ gives the flat plateau; the outer face sits 5 dB above the inner one because $M_{A1} S \propto a$. Above $ka = 1$ the load is the resistance $\rho c/S$, so $p_{front} \approx \rho c u$ for either face (the same 0.59 Pa/N at 10 kHz in the stiff case) and it falls with the velocity. In the far field with constant force, $\omega \rho S u$ is flat in the mass-controlled region ($\omega \propto 1/\omega$), the familiar flat loudspeaker response above resonance. The stiff link is one 11 g piston, flat to 10 kHz in this lumped model (a real cone would break up; the model has no cone modes).

The dual-diaphragm principle (soft link). Up to about 1 kHz the outer cone provides 78 % of the volume velocity ($S_o/(S_i + S_o)$). Around 1.6 kHz the link resonance boosts the output (+8 dB hump, the outer cone resonating on C_{md}). Above about 2 kHz the outer cone decouples, its velocity falls 12 dB per octave faster and ends up in anti-phase, so around 3 kHz the two partially cancel (dip, Fig. 14). From about 4 kHz upward only the inner cone radiates: –13 dB relative to the low-frequency plateau ($= 20 \log(60/270)$) but *flat*, extending the bandwidth well beyond where a 210 cm² cone of 11 g would have rolled off or broken up. The design intent is large area for bass efficiency, small light area for treble, and the soft joint is a mechanical crossover between them.

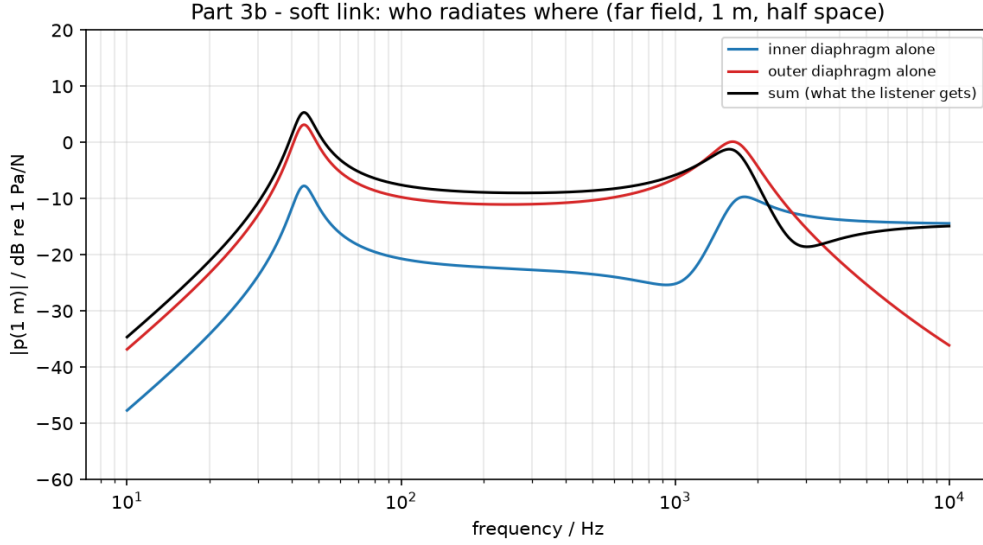


Figure 14: Part 3b: far-field contribution of each diaphragm and their sum, soft link — the 3 kHz cancellation is where the decoupled outer cone has swung into anti-phase.

Modelling choices and validity. As instructed, each diaphragm is loaded as an independent baffled piston (no mutual radiation impedance between the two, no dipole cancellation of the inner cone even though it sits in the throat of the outer one) and the total volume velocity is the plain sum $U_i + U_o$. Both overestimate the treble output somewhat. The lumped radiation network is itself a fit that is trustworthy up to ka of a few, i.e. about 2–5 kHz for the outer cone, so the top octave of the soft-link curves is qualitative.

4 Part 4 — Coupled electrical–mechanical system

4.1 Model

The coil is driven by an ideal 1 V source through R_c and L_e . The current produces the Lorentz force $f = Bl i$ on the coil mass M_{mc} ; the coil velocity induces the back-EMF $v = Bl u_c$ in the electrical loop. M_{mc} is joined to M_{m1} by $C_{ms} \parallel R_{ms}$, and M_{m1} to ground by $C_{ms2} \parallel R_{ms2}$.

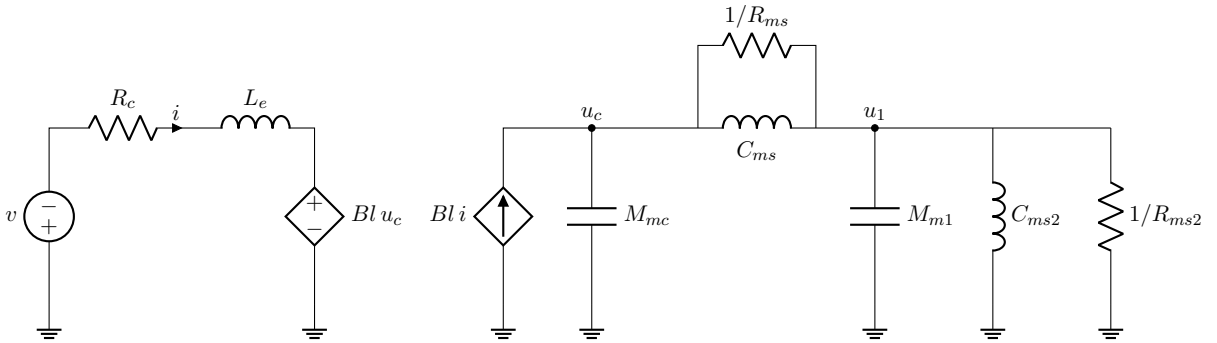


Figure 15: Part 4: electrical loop (left) coupled to the mobility circuit (right) by the gyrator pair $f = Bl i$ (VCCS, sensing i as the voltage across R_c) and $v = Bl u_c$ (VCVS).

Physical	Value	SPICE element	Value
Bl	$1 \text{ T} \times 1.5 \text{ m} = 1.5 \text{ T m}$	VCCS gain Bl/R_c , VCVS gain Bl	3, 1.5
R_c, L_e	$0.5, 10 \text{ }\mu\text{H}$	R, L	$0.5, 10 \text{ }\mu\text{H}$
M_{mc}	5 g	C (u_c -gnd)	5 mF
C_{ms}, R_{ms}	$0.01 \text{ mm/N}, 1 \text{ N s/m}$	L, R (u_c - u_1)	$10 \text{ }\mu\text{H}, 1$
M_{m1}	100 g	C (u_1 -gnd)	100 mF
C_{ms2}, R_{ms2}	$10 \text{ mm/N}, 0.1 \text{ N s/m}$	L, R (u_1 -gnd)	10 mH, 10

Table 13: Part 4 element values.

Hand estimates. Electrical damping from the coil: $(Bl)^2/R_c = 4.5 \text{ N s/m}$, larger than any mechanical damper. Heavy mass on its suspension: $f \approx 1/(2\pi\sqrt{M_{m1}C_{ms2}}) = 5.0 \text{ Hz}$. Coil on its stiff spring against the (almost fixed) heavy mass: $f \approx 1/(2\pi\sqrt{\mu C_{ms}})$ with the reduced mass $\mu = M_{mc}M_{m1}/(M_{mc} + M_{m1}) = 4.76 \text{ g}$, giving 729 Hz. The coil inductance takes over the electrical impedance above $f = R_c/(2\pi L_e) \approx 8.0 \text{ kHz}$. The electrical input impedance is

$$Z_E = R_c + j\omega L_e + \frac{(Bl)^2}{Z_M},$$

so mechanical resonances (minima of Z_M) show up as *maxima* of $|Z_E|$ — the motional impedance is the inverted mechanical impedance.

4.2 Results and discussion

Sign check of the back-EMF. The lab warns about the signs of the dependent generators, so the polarity was tested explicitly (Table 14). With the VCVS as drawn the electrical damping $(Bl)^2/R_c$ cuts the 4.9 Hz velocity peak by a factor 45 and the current at resonance from 2 A to 44 mA: the back-EMF opposes the drive, the physically correct sign. Flipping it removes the low peak and grows the 724 Hz one instead (negative damping). Power check: the electrical power into the VCVS, $v i = Bl u_c i$, equals the mechanical power $f u_c = Bl i u_c$ delivered by the VCCS, so the transducer is lossless as it must be.

VCVS gain	max $ u_c $ [m/(s V)]	at	$ Z_E $ there	DC current
+1.5 (as drawn)	0.652	4.9 Hz	22.6	1.915 A
0 (no back-EMF)	29.5	4.9 Hz	0.50	2.000 A
-1.5 (flipped)	0.889	724 Hz	1.47	1.922 A

Table 14: Part 4 sign check of the back-EMF source.

	u_c (coil) [m/(s V)]	u_1 (mass M_{m1})
1 Hz	0.188	0.188
peak	4.90 Hz, 0.652	4.90 Hz, 0.652
dip	158.5 Hz, 4.3×10^{-4}	436.5 Hz, 0.0158
peak	732.8 Hz, 0.537	716 Hz, 0.0272
10 kHz	6.0×10^{-3}	1.8×10^{-6}

Table 15: Part 4a extracted numbers; $|u_1/u_c| = 1.00$ below 20 Hz, 1.65 at 100 Hz, 0.055 at 700 Hz, 3×10^{-4} at 10 kHz.

a) Velocities (Fig. 16). *4.9 Hz, rigid-body mode.* C_{ms} (0.01 mm/N) is a thousand times stiffer than C_{ms2} , so at low frequency coil and mass move as one lump of 105 g on the soft suspension C_{ms2} (hand: 4.91 Hz); both velocities are equal and in phase. The peak is broad because the electrical damping of 4.5 N s/m dominates ($Q \approx \omega_0 M/R \approx 0.7$); with the back-EMF removed the same mode

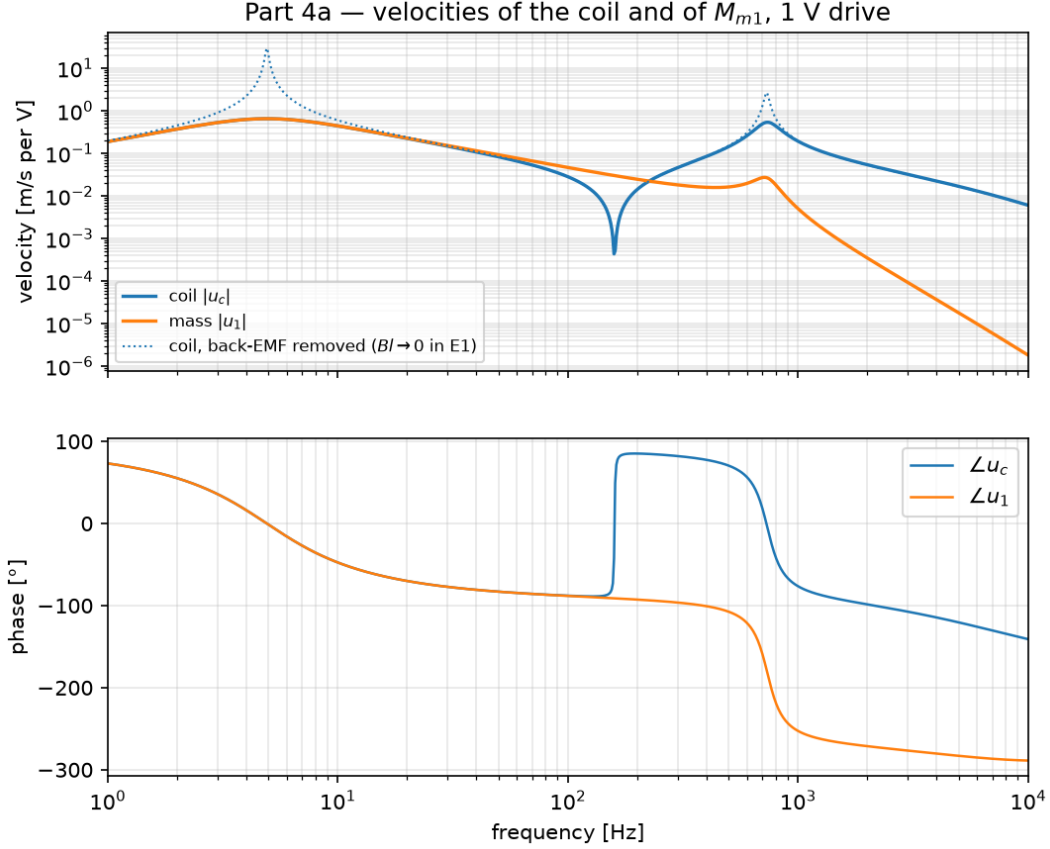


Figure 16: Part 4a: coil velocity u_c and mass velocity u_1 per volt, magnitude and phase, 1 Hz to 10 000 Hz; dotted: u_c with the back-EMF removed.

is damped only by $R_{ms2} = 0.1 \text{ N s/m}$ and peaks at 29.5 m/(s V) . Below the resonance the motion is compliance-controlled ($u \propto j\omega$, $+90^\circ$), between the resonances mass-controlled ($u \propto 1/j\omega$, -90°).

158.5 Hz, the coil stops. This is the anti-resonance of the mechanical load: M_{m1} on the stiff spring C_{ms} resonates ($1/(2\pi\sqrt{M_{m1}C_{ms}}) = 159 \text{ Hz}$) and presents an almost infinite impedance to the coil, so u_c drops to $4 \times 10^{-4} \text{ m/(s V)}$ while u_1 keeps moving ($|u_1/u_c|$ about 40 right at the dip). It is the tuned-mass-damper effect; the phase of u_c jumps by 180° there.

733 Hz, coil-on-spring mode. The 5 g coil bounces on C_{ms} against the heavy, now almost stationary mass (reduced-mass estimate 729 Hz). u_c peaks at 0.54 m/(s V) ; u_1 shows only a small hump (0.027) and then falls at 40 dB per decade because the spring cannot transmit the force to M_{m1} at high frequency (mass isolation). Above 733 Hz u_c falls as $1/(\omega M_{mc})$, and above 8 kHz L_e also starts to limit the current. The shallow minimum of u_1 at 436 Hz is a zero of its transfer function (the spring force and the C_{ms2}/R_{ms2} branch partially cancel), not a system resonance.

b) Mechanical input impedance seen by the force (Fig. 17a). $Z_M = f/u_c = Bl I(V_{s1})/V(u_c)$ (KiCad expression $1.5 \cdot I(V_{s1})/V(/u_c)$).

Feature	f	$ Z_M \text{ [N s/m]}$	Phase
1 Hz	—	15.2	-90° (C_{ms2} dominates)
resonance (minimum)	4.90 Hz	0.10 ($\approx R_{ms2}$)	0°
anti-resonance (maximum)	158.5 Hz	6.9×10^3	0° (flip $+90 \rightarrow -90$)
resonance (minimum)	732.8 Hz	1.13 ($\approx R_{ms}$)	0°
10 kHz	—	$313 \approx \omega M_{mc}$	$+90^\circ$ (coil mass)

Table 16: Part 4b extracted numbers.

The impedance seen by the force is that of the pure mechanical network: the transducer is lossless, so Z_M does *not* contain the electrical damping, which acts on the electrical side. Minima are where the force meets almost no impedance, only the dampers, i.e. the two resonances; at 4.9 Hz $|Z_M| \approx R_{ms2}$ and at 733 Hz $|Z_M| \approx R_{ms}$, so the damper values can be read straight off the plot. The maximum is the anti-resonance where M_{m1} on C_{ms} takes all the force and the coil cannot move. The phase sequence $-90^\circ / +90^\circ / -90^\circ / +90^\circ$ reads: compliance-like below 4.9 Hz (C_{ms2}), mass-like between 4.9 and 158 Hz (the 105 g lump), compliance-like between 158 and 733 Hz (C_{ms} , the coil is spring-controlled), mass-like above (M_{mc}).

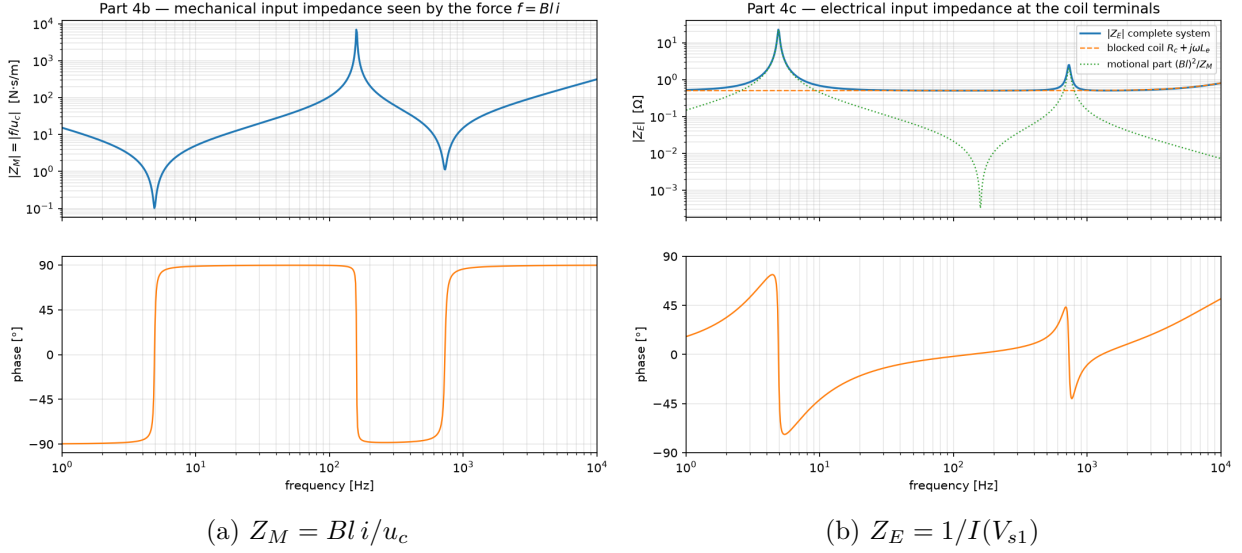


Figure 17: Part 4b/c: mechanical impedance seen by the electromagnetic force and electrical impedance seen by the 1 V source — mirror images scaled by $(Bl)^2$.

Feature	f	$ Z_E $
1 Hz	—	0.522
peak	4.90 Hz	22.6 (motional part 22.1)
broad minimum	100–400 Hz	$0.500 = R_c$
peak	732.8 Hz	2.48 (motional 1.98)
dip	1.4 kHz	0.5025
10 kHz	—	$0.797 \approx R_c + j\omega L_e = 0.803$

Table 17: Part 4c extracted numbers.

c) Electrical input impedance (Fig. 17b). $Z_E = R_c + j\omega L_e + (Bl)^2/Z_M$: the motional term is the *inverse* of the mechanical impedance (Lecture 4A, impedance conversion $Z_{E,M} = (Bl)^2 Y_M$). So the mechanical *resonances* (Z_M minima) appear as *peaks* in $|Z_E|$, $(Bl)^2/R_{ms2} = 22.5 \Omega$ at 4.9 Hz and $(Bl)^2/1.13 = 2.0 \Omega$ at 733 Hz on top of R_c , while the mechanical *anti-resonance* (158 Hz) is where Z_E is closest to R_c ($(Bl)^2/6.9 \times 10^3 = 0.3 \text{ m}\Omega$): the coil is blocked, there is no back-EMF, and the electrical side sees the blocked-coil impedance. The Z_E peaks are sharp because on the electrical side the mechanical resonance is loaded only by the mechanical dampers ($Q \approx 2\pi \cdot 4.9 \cdot 0.105/0.1 \approx 32$ for the first mode). The phase of Z_E swings about $\pm 75^\circ$ around the peaks (inductive below a resonance, capacitive above): a motional impedance looks like a parallel RLC on the electrical side, the classic loudspeaker impedance curve. Above about 1.4 kHz the motional term is negligible and Z_E follows $R_c + j\omega L_e$ with its corner at 8 kHz ($+47^\circ$, 0.8 at 10 kHz). Compared with b), Z_M and Z_E are mirror images on a log scale, every minimum of one a maximum of the other, scaled by $(Bl)^2 = 2.25$.

f	u_c	u_1	Z_M	Z_E	Physical origin
4.9 Hz	peak 0.65	peak 0.65	min 0.10 ($\approx R_{ms2}$)	peak 22.6	both masses (105 g) on the soft suspension C_{ms2} ; rigid-body mode, damped mostly by $(Bl)^2/R_c$
158.5 Hz	dip 4×10^{-4} (coil stands still)	still moving	max 6.9×10^3	$\approx R_c$	M_{m1} resonating on the stiff spring C_{ms} absorbs the force: anti-resonance, tuned-mass-damper effect
436 Hz	—	shallow dip	—	—	zero of the u_1 transfer function (spring force vs. suspension branch), not a system mode
733 Hz	peak 0.54	small hump 0.027	min 1.13 ($\approx R_{ms}$)	peak 2.5	coil (5 g) bouncing on C_{ms} against the nearly stationary M_{m1} (reduced mass 4.76 g, 729 Hz)
8 kHz	roll-off steepens	—	—	inductive rise	electrical corner $R_c/(2\pi L_e)$; L_e limits the current, no mechanical origin

Table 18: Part 4d: all features of the three plots and their origin.

d) Every resonance and anti-resonance (Fig. 18). Forces as a function of the driving current: the coil force is always $f = Bli$; the force in the spring C_{ms} is $f - j\omega M_{mc}u_c$, which is $\approx f$ below 158 Hz (the coil mass is negligible) and tends to zero above 733 Hz, where the coil mass takes the whole force.

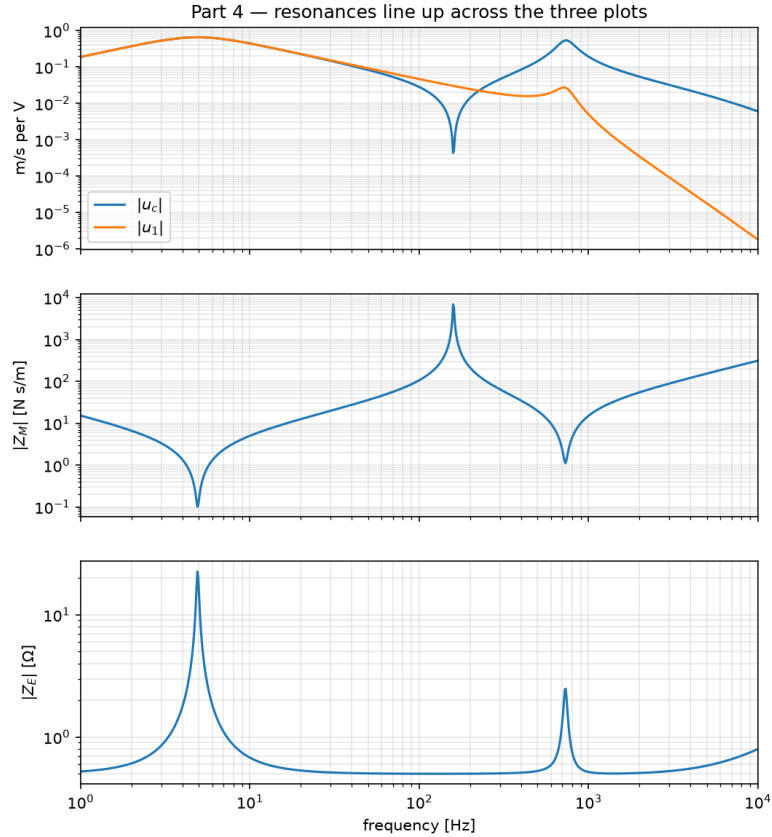


Figure 18: Part 4d: velocities, Z_M and Z_E on one frequency axis; every resonance of one plot lines up with an anti-resonance of another.

Modelling choices and validity. The mobility analogy was used for the mechanical side; in the impedance analogy the same circuit needs a gyrator (H/F sources) instead of the G/E pair, and the plots are identical. The coil current is sensed as the voltage across R_c (gain Bl/R_c), which is exact for a linear resistor. Frequencies are read from a 200-points-per-decade sweep, accurate to about 0.5 %.

5 Conclusion

The four systems reduce to the same two ideas. First, an electromechanoacoustic system is an RLC network once the analogy is chosen consistently: in the mobility analogy every mass is a capacitor to ground and every spring an inductor, in the acoustic impedance analogy every closed volume is a grounded capacitor and every narrow tube an inductor with its loss in series. Second, a transducer is a pair of controlled sources with a constant gain (S or Bl), and everything on the far side of it is seen through that gain: an acoustic load becomes $S^2 Z_A$ on the mechanical side, a mechanical impedance becomes $(Bl)^2/Z_M$ on the electrical side, which is why mechanical resonances show up as peaks in the electrical impedance.

The simulations reproduced every hand estimate: the 51 Hz single-mass resonance and the 1.3 kHz absorber of the Lowther dual cone (Part 1); the ladder resonances of the silencer and the fact that a pressure source and a volume-velocity source excite complementary sets of natural frequencies (Part 2); the 4 g of air mass and the 22 N s/m of radiation resistance the baffle adds to the two cones (Part 3); the 4.9 Hz and 733 Hz modes of the coil system and their mirror image in Z_E (Part 4). The most instructive results are the ones the circuit gives for free: the mechanical crossover of the dual diaphragm, whose far-field response stays flat to 10 kHz at the cost of 13 dB once only the inner cone radiates; the silencer amplifying the exhaust by about 40 dB at its resonances while attenuating 110 dB at 1 kHz, of which the upper half is outside the validity of the lumped model; and the sign of the back-EMF, which is easy to get wrong and immediately visible as a 45° change in the velocity peak.

Appendix A: reproducing the simulations

Everything lives in 5. Semester/Electroacoustics/Labs/Lab A/:

- KiCad/ holds one project per sheet (Table 19), each a full KiCad 10 project: `.kicad_pro`, `.kicad_sch` with the `.ac` directive as sheet text, `.wbk` workbook with the traces preloaded, and the exported `.cir`. Open the schematic and run *Inspect* → *Simulator*.
- `sim/part1.py` to `part4.py` — standalone scripts that draw the sheets (`schdraw`), export the netlist, run ngspice in batch, post-process with numpy and write the figures; helper modules alongside.
- `results/partN.md` — element tables, all extracted numbers and the interpretation behind this report.

Project folder	Part	Drive / variant
Part1_DualDiaphragm_Stiff	1	1 N, $C_{md} = 10^{-10}$ m/N
Part1_DualDiaphragm_Soft	1	1 N, $C_{md} = 3 \times 10^{-6}$ m/N
Part2a_Silencer_PressureSource	2a	1 Pa, ideal open end
Part2b_Silencer_VolumeVelocitySource	2b, 2c	1 m ³ /s, ideal open end
Part2d_Silencer_Radiation	2d, 2e	1 m ³ /s, radiation network
Part3_Coupled_Stiff / _Soft	3	Part 1 + radiation on both diaphragms
Part4_Coil_Electromechanical	4	1 V

Table 19: KiCad projects.

The batch path used by the scripts is

```

kicad-cli sch export netlist --format spice -o X.cir X.kicad_sch
ngspice -b run.cir
# run.cir:  .include X.cir
#           .control
#           run
#           wrdata out.txt v(/node) i(Vs1) ...
#           .endc

```

and `wrdata` writes frequency, real and imaginary part per vector.

Translation to LTspice for the quiz. The circuits are tool-agnostic. KiCad’s `GSOURCE` (VCCS) is LTspice’s `G` and `ESOURCE` (VCVS) is `E`; the SPICE line is identical (`G1 N+ N- C+ C- gain`, current flows from `N+` through the source to `N-`). A `0V` voltage source in series with a branch gives its current as `I(Vs)` in both tools. The mobility-analogy impedance trick is the same: drive with a `1 A` current source and plot `1/V(u)`; in LTspice use *Add traces* with the expression, in KiCad a user-defined signal. Every node needs a DC path to ground, so a current source feeding a capacitor-only branch gets a leak resistor — chosen far above the largest impedance in the circuit (`1 T` for the silencer, whose input impedance reaches `1.5 G`), or it becomes part of the physics (Part 2b).

Appendix B: KiCad schematics

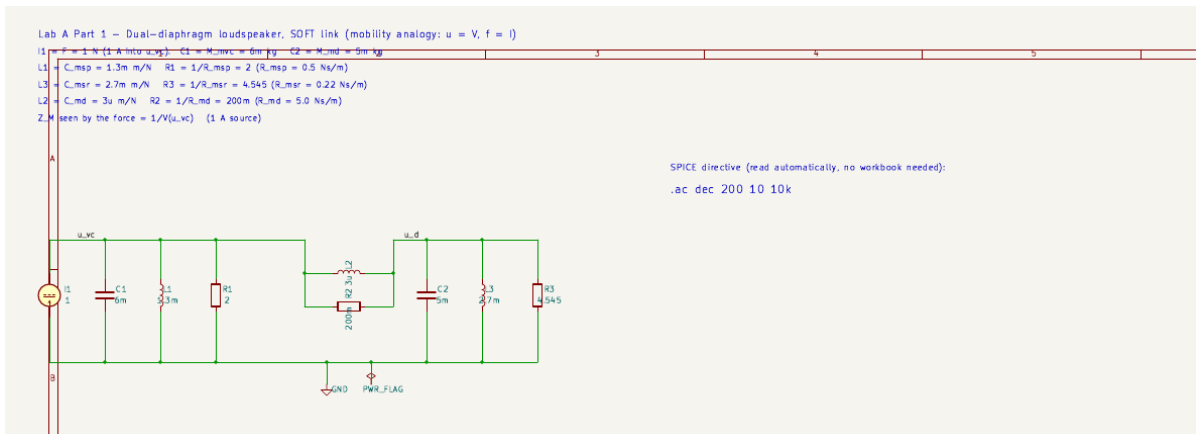


Figure 19: Part 1, soft link (the stiff sheet differs only in L_2 and R_2).

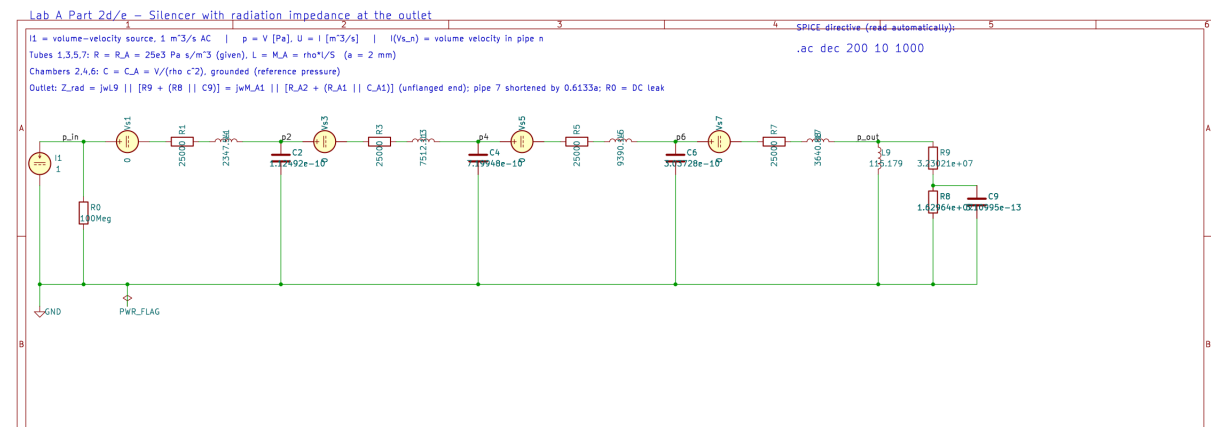


Figure 20: Part 2d: silencer with the radiation network at the outlet (sheets 2a/2b are the same ladder with an ideal open end and a pressure or volume-velocity source).

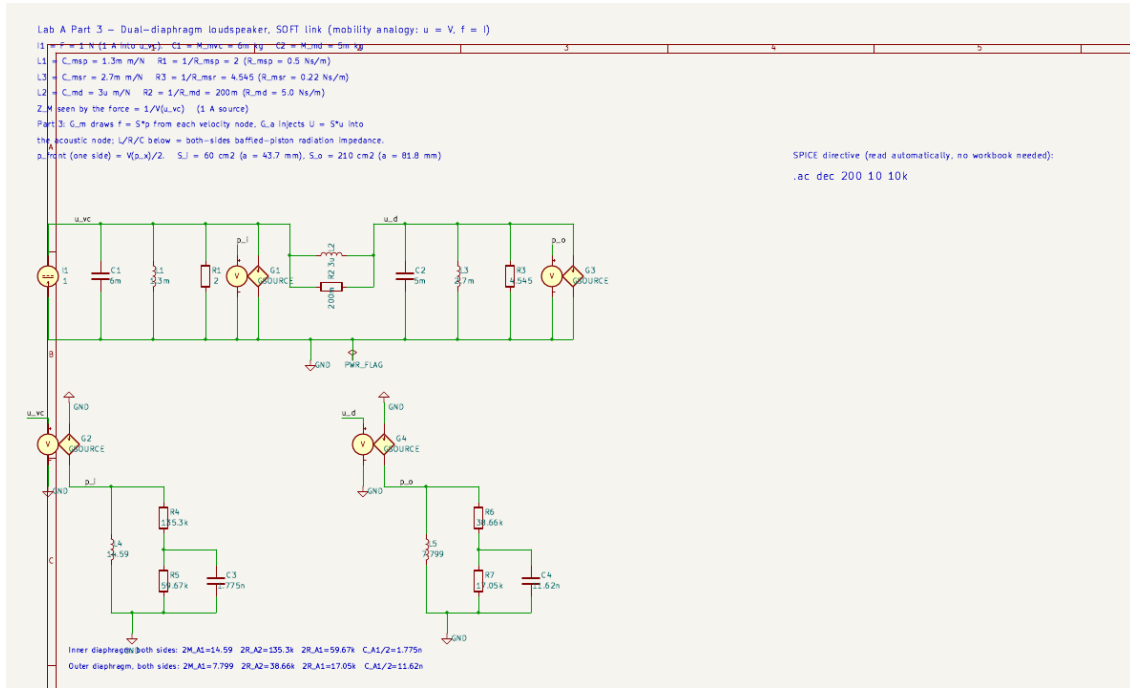


Figure 21: Part 3, soft link: the Part 1 network with two radiation sub-networks coupled by G-sources.

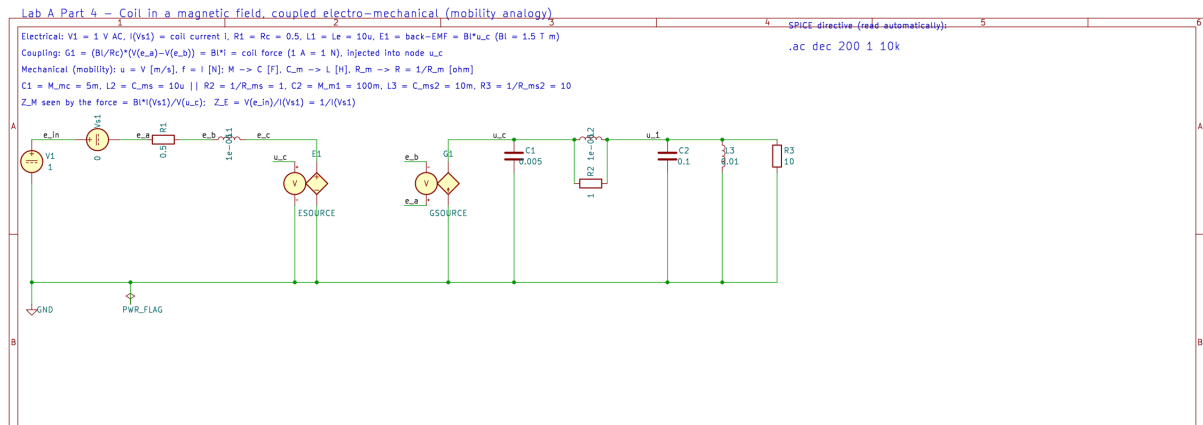


Figure 22: Part 4: electrical loop with back-EMF VCVS, force VCCS into the mobility network.